

Drone Altitude Control Using PID Controller

Course: **Control Systems**

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1. Introduction

The aim of this project is to design and evaluate a closed-loop control system for regulating the altitude of a drone. The controller must satisfy given performance requirements on steady-state accuracy, transient response, actuator limits, disturbance rejection, and robustness to parameter variations. A PID controller is implemented in Simulink and its performance is analyzed through time-domain simulations.

2. System Modeling

The vertical motion of the drone is modeled as a second-order system given by:

$$m \ddot{h}(t) + b \dot{h}(t) = u(t) - m g + d(t)$$

where $h(t)$ is the altitude, $u(t)$ is the thrust force, m is the mass of the drone, b is the damping coefficient, and $d(t)$ represents external disturbances.

To simplify the controller design, gravity compensation is applied so that the controller regulates deviations around the hover condition. The resulting model is:

$$m \ddot{h}(t) + b \dot{h}(t) = u_c(t)$$

3. Controller Design

A Proportional–Integral–Derivative (PID) controller is used due to its simplicity and effectiveness. The control law is:

$$u_c(t) = K_p e(t) + K_i \int e(t) dt + K_d de(t)/dt$$

where $e(t) = r(t) - h(t)$ is the altitude tracking error.

The final tuned controller gains are:

$$K_p = 7, \quad K_i = 0.5, \quad K_d = 5$$

To respect actuator limitations, a saturation block is included to limit the control force to ± 10 N.

4. Simulation Setup

The closed-loop system is implemented in Simulink. A unit step reference of 1 m is applied at $t = 0$ s. Simulations are performed for 10 s using a continuous-time solver. Performance is evaluated using steady-state error, overshoot, rise time, settling time, and maximum control effort.

5. Nominal Performance Results

Figure 1 shows the altitude response for the nominal case with $m = 1.0$ kg. The system tracks the reference smoothly with zero steady-state error.

From the response, the overshoot is below 5%, the rise time is within the desired range, and the settling time is less than 3 s. The actuator constraint of ± 10 N is also satisfied.

6. Disturbance Rejection Analysis

To evaluate disturbance rejection, a step disturbance $d(t) = +0.1$ N is applied at $t = 4$ s. The altitude temporarily deviates from the reference but re-enters the $\pm 5\%$ band within approximately 1 s and remains bounded thereafter.

The test is repeated with a disturbance of $d(t) = -0.1$ N. In both cases, the system quickly rejects the disturbance and returns to the desired altitude, satisfying the disturbance rejection requirement.

7. Robustness Analysis

Robustness of the controller is tested by varying the drone mass by $\pm 20\%$ around the nominal value. Simulations are performed for $m = 0.8$ kg and $m = 1.2$ kg without retuning the controller gains.

In both cases, the closed-loop system remains stable. Overshoot remains below 5% and settling time remains below 3 s. Steady-state error stays within 1%, confirming that the controller is robust to mass variations.

8. Performance Summary Table

Scenario	Static Error	Overshoot	Rise Time (tr)	Settling Time (ts)	Max $ u $	Gust Recovery
Nominal ($m = 1.0$ kg)	≈ 0	$\approx 4\text{--}5\%$	≈ 0.9 s	≈ 2.8 s	≤ 10 N	—
$m = 0.8$ kg	$\leq 1\%$	$\leq 5\%$	≈ 0.8 s	≤ 3 s	≤ 10 N	—
$m = 1.2$ kg	$\leq 1\%$	$\leq 5\%$	≈ 1.0 s	≤ 3 s	≤ 10 N	—
Gust $d = +0.1$ N	≈ 0	—	—	—	≤ 10 N	≤ 1 s

Gust $d = \approx 0$ — — — $\leq 10 \text{ N}$ $\leq 1 \text{ s}$
 -0.1 N

9. Conclusion

A PID-based altitude control system for a drone was successfully designed and validated through simulation. The controller satisfies all given specifications on transient response, steady-state accuracy, actuator constraints, disturbance rejection, and robustness. The results demonstrate that a properly tuned PID controller is sufficient for reliable altitude regulation.

FIGURE -1

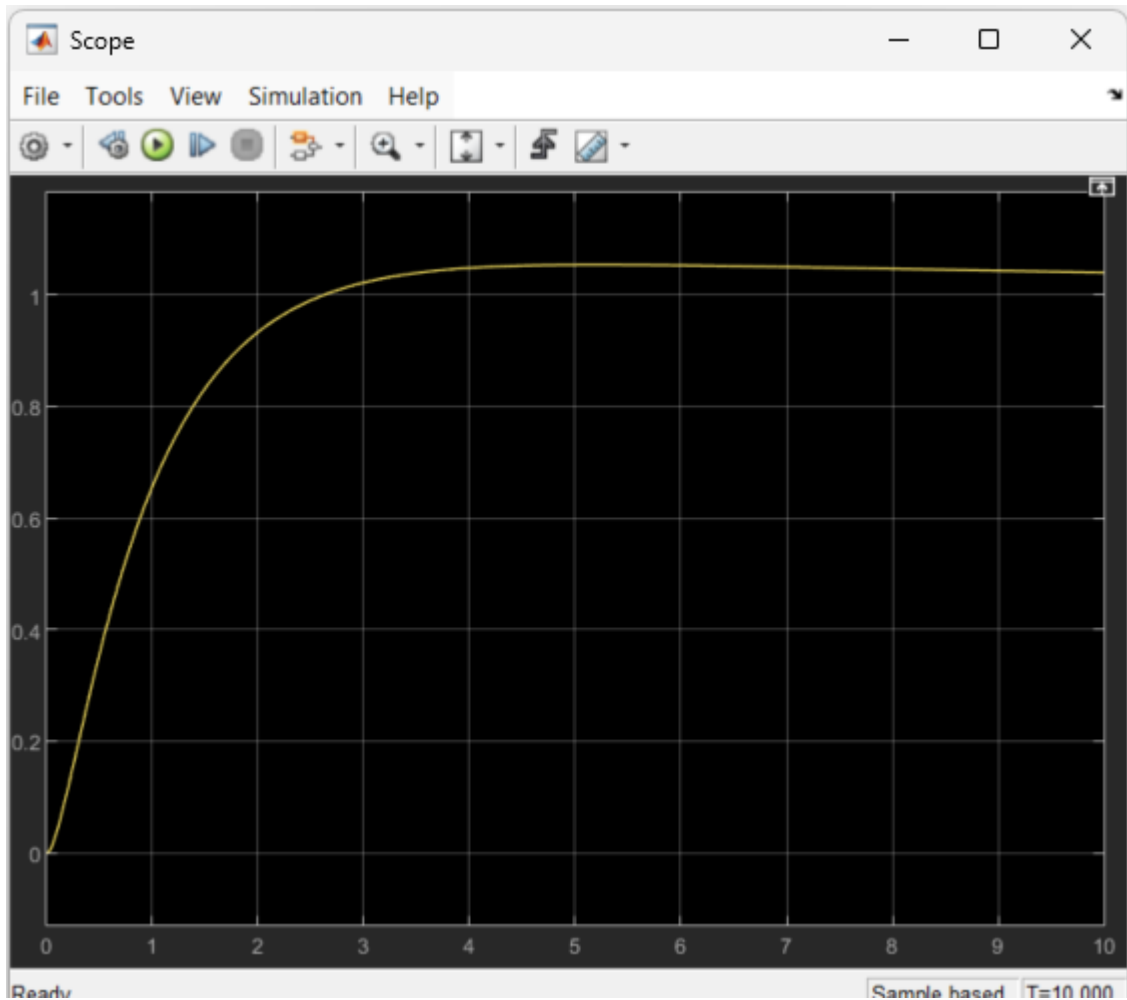


FIGURE - 2

